Name of Student	Roll No.	Date:

Experiment No. 8

Tiltle: Design of SRAM Cell Layout.

Pre-lab Requisites:

1. Knowledge of Pass transistor logic
2. Knowledge of one bit memory
3. Knowledge of SRAM cell

Objectives:

1. SRAM Basics
2. CMOS SRAM Cell
3. CMOS SRAM Cell Design
4. READ Operation
5. WRITE Operation

Equipments:

Sr. No.	Description	Specification
1	System	Processor - 2GHz or Higher, RAM – 2GB or Higher, OS – Windows (32 Bit with SP1 or Higher)
2.	EDA tool	Microwind

Theory:

A] SRAM Basics

The memory circuit is said to be static if the stored data can be retained indefinitely, as long as the power supply is on, without any need for periodic refresh operation. The data storage cell, i.e., the one-bit memory cell in the static RAM arrays, invariably consists of a simple latch circuit with two stable operating points. Depending on the preserved state of the two inverter latch circuit, the data being held in the memory cell will be interpreted either as logic **'0'** or as logic **'1'**. To access the data contained in the memory cell via a bit line, we need at least one switch, which is controlled by the corresponding word line as shown in Fig 7.1

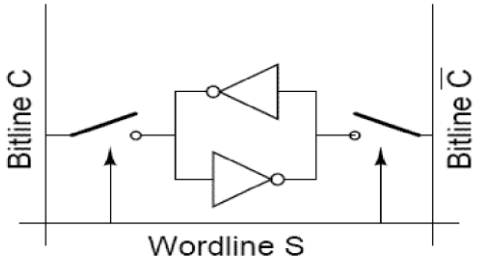


Fig. 7.1 SRAM Cell

B] CMOS SRAM Cell

A low power SRAM cell may be designed by using cross-coupled CMOS inverters. The most important advantage of this circuit topology is that the static power dissipation is very small; essentially, it is limited by small leakage current. Other advantages of this design are high noise immunity due to larger noise margins, and the ability to operate at lower power supply voltage. The major disadvantage of this topology is larger cell size. The circuit structure of the full CMOS static RAM cell is shown in Figure 7.2. The memory cell consists of simple CMOS inverters connected back to back, and two access transistors. The access transistors are turned on whenever a word line is activated for read or write operation, connecting the cell to the complementary bit line columns.

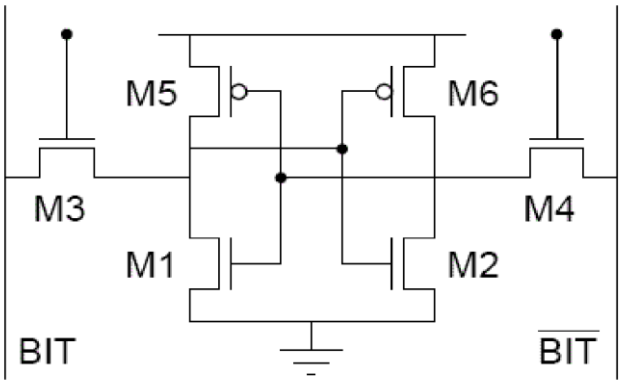


Fig. 7.2 Full CMOS SRAM Cell

C] CMOS SRAM Cell Design

To determine **W/L** ratios of the transistors, a number of design criteria must be taken into consideration. The two basic requirements, which dictate **W/L** ratios, are that the data **read** operation should not destroy the stored information in the cell. The cell should allow stored information modification during **write** operation. In order to consider operations of SRAM, we have to take into account, the relatively large parasitic column capacitance and column pull-up transistors as shown in Figure .When none of the word lines is selected, the pass transistors M3 and M4 are turned off and the data is retained in all memory cells. The column capacitances are charged by the pull-up transistors P1 and P2. The voltages across the column capacitors reach **VDD - VT**.

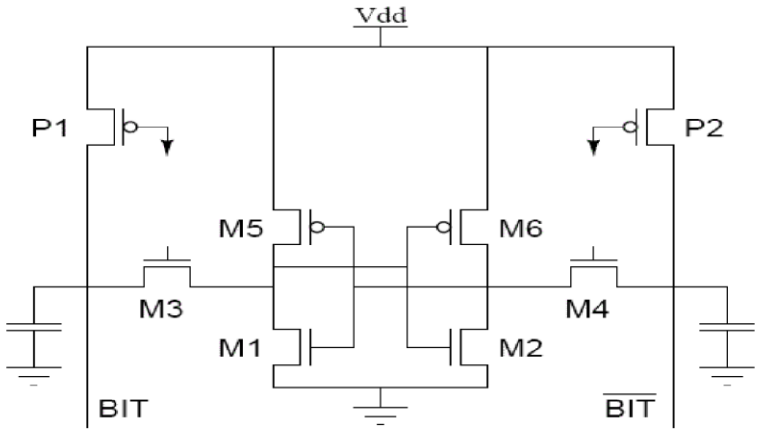


Fig. 7.3 CMOS SRAM cell with precharge transistors

READ Operation:

Consider a data read operation, shown in Figure 7.4, assuming that logic **'0'** is stored in the cell. The transistors M2 and M5 are turned off, while the transistors M1 and M6 operate in linear mode. Thus internal node voltages are **V1 = 0** and **V2 = VDD** before the cell access transistors are turned on. The active transistors at the beginning of data read operation are shown in Figure

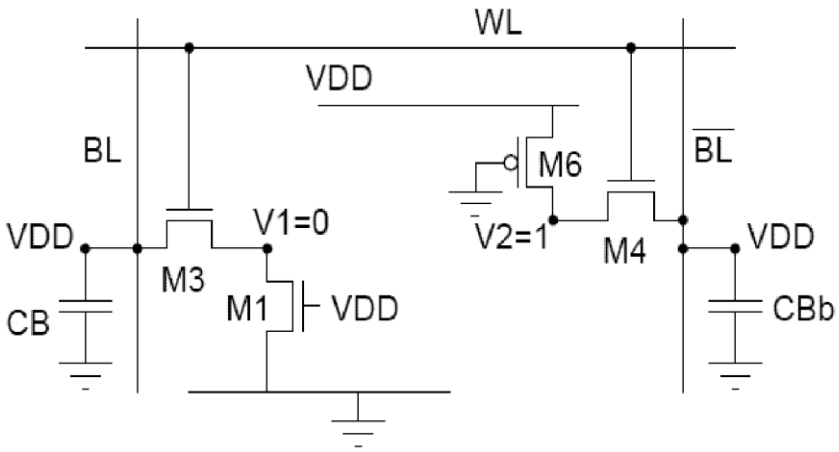


Fig. 7.4 Read Operation

After the pass transistors M3 and M4 are turned on by the row selection circuitry, the voltage **CBb** of will not change any significant variation since no current flows through M4. On the other hand M1 and M3 will conduct a nonzero current and the voltage level of **CB** will begin to drop slightly. The node voltage **V1** will increase from its initial value of **'0'V.** The node voltage **V1** may exceed the threshold voltage of M2 during this process, forcing an unintended change of the stored state. Therefore voltage must not exceed the threshold voltage of M2, so the transistor M2 remains turned off during read phase, i.e.,

$$V_{1max} \leq V_{T,2} \quad \text{- (1)}$$

The transistor M3 is in saturation whereas M1 is linear, equating the current equations we get

$$\frac{\beta_{N3}}{2}(V_{DD} - V_1 - V_{TN}) = \frac{\beta_{N1}}{2}(2(V_{DD} - V_{TN})V_1 - {V_1}^2) \quad \textbf{- (2)}$$

Substituting **Eq 1** in **Eq 2** we get

$$\frac{\beta_{N3}}{\beta_{N1}} = \frac{(W/L)_3}{(W/L)_1} \leq 2V_{TN}\frac{(V_{DD} - 1.5V_{TN})}{(V_{DD} - 1.5V_{TN})^2} \quad \textbf{- (3)}$$

WRITE Operation

Consider the write **'0'** operation assuming that logic **'1'** is stored in the SRAM cell initially. Figure 7.5 shows the voltage levels in the CMOS SRAM cell at the beginning of the data write operation. The transistors M1 and M6 are turned off, while M2 and M5 are operating in the linear mode. Thus the internal node voltage **V1 = VDD** and **V2 = 0** before the access transistors are turned on. The column voltage **Vb** is forced to **'0'** by the write circuitry. Once M3 and M4 are turned on, we expect the nodal voltage **V2** to remain below the threshold voltage of M1, since M2 and M4 are designed according to **Eq. 1**.

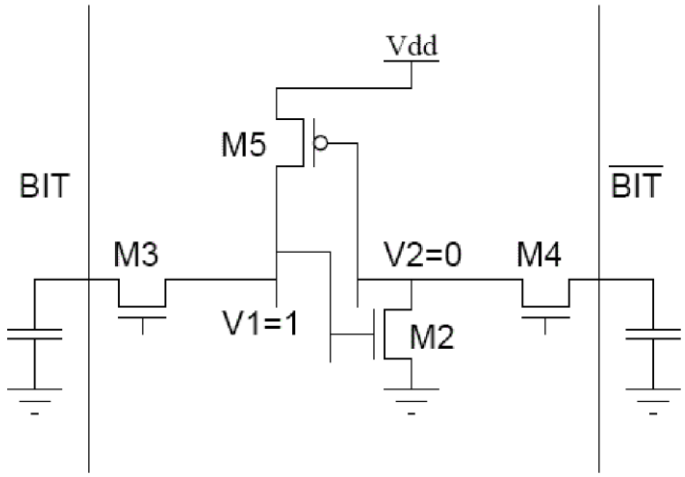


Fig. 7.5 SRAM start of write '0'

The voltage at node 2 would not be sufficient to turn on M1. To change the stored information, i.e., to force **V1 = 0** and **V2 = VDD**, the node voltage **V1** must be reduced below the threshold voltage of M2, so that M2 **turns off.** When the transistor M3 operates in linear region while M5 operates in saturation region. Equating their current equations we get

$$\frac{\beta_{P5}}{2}(0 - V_{DD} - V_{TP})^2 = \frac{\beta_{N3}}{2}(2(V_{DD} - V_{TN})V_1 - {V_1}^2) \quad \textbf{- (4)}$$

Rearranging the condition of in the result we get

$$\frac{\beta_{P5}}{\beta_{N3}} = \frac{\mu_p}{\mu_n}\frac{(W/L)_5}{(W/L)_3} \leq \frac{2((V_{DD} - 1.5V_{TN})V_{TN})}{(V_{DD} - V_{TP})} \quad \textbf{- (5)}$$

WRITE Circuit

The principle of write circuit is to assert voltage of one of the columns to a low level. This can be achieved by connecting either or to ground through transistor M3 and either of M2 or M1. The transistor M3 is driven by the column decoder selecting the specified column. The transistor M1 is on only in the presence of the write enable signal and when the data bit to be written is **'0'**. The transistor M2 is on only in the presence of the write signal and when the data bit to be written is **'1'**. The circuit for write operation is shown in Figure

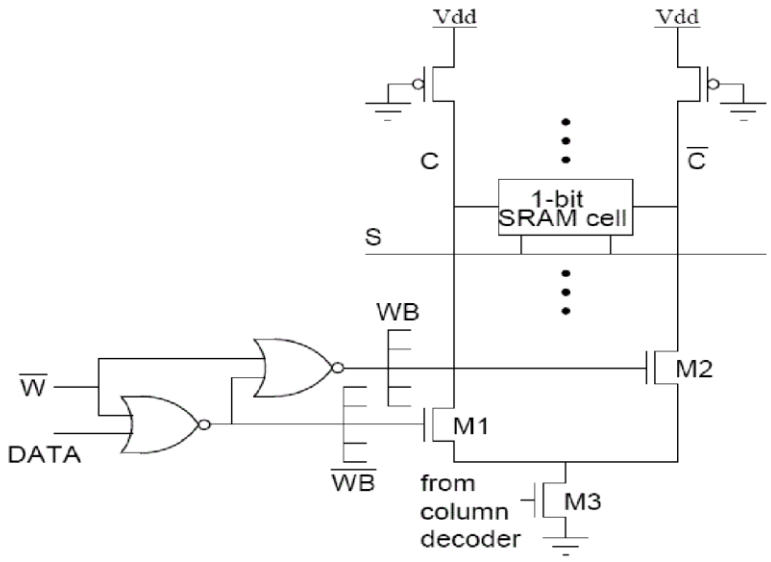
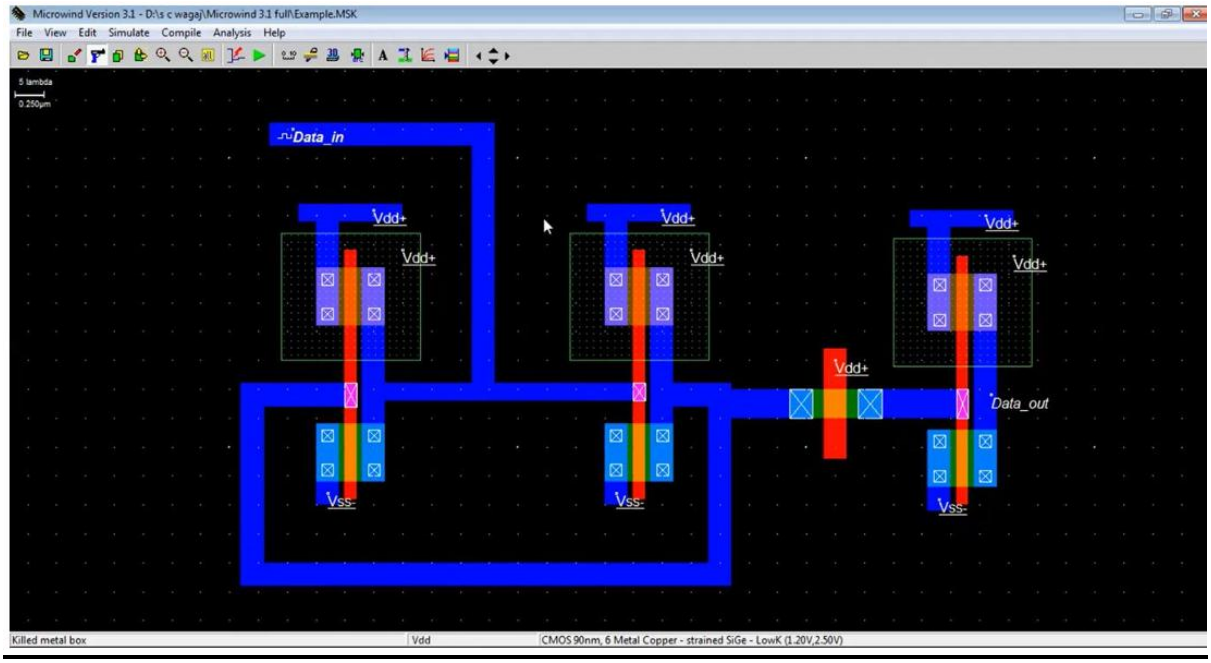
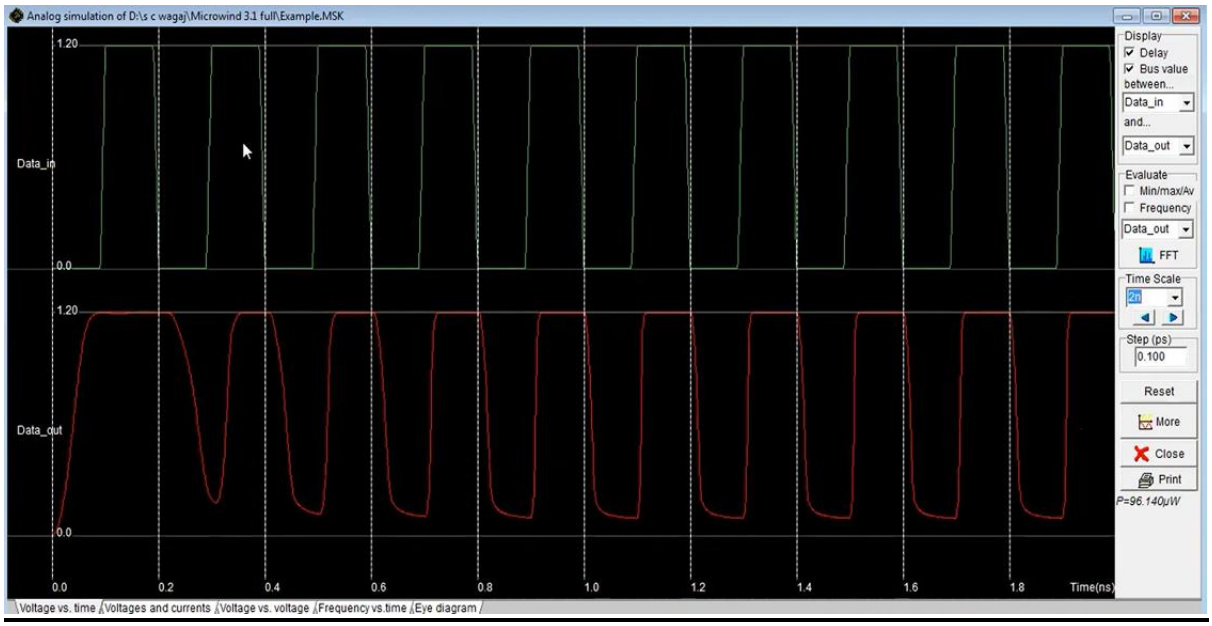


Fig. 7.6 Circuit for write operation

SRAM Cell Layout:



Simulation waveform:



Conclusion:

Checked By:

Name of Subject Teacher	Sign with Date